

SNAPFRAME · BUILD PLAN

SnapFrame prototype build plan

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-02	Amish Chadha	Node polymer changed to glass- or carbon-filled PA12-class nylon (decided by Amish on 2026-10-02); printer and fume notes to match
0.1	2026-10-01	Amish Chadha	First build plan, from the template with pictures by component and step; design made constructable (SNF-DDR-003)

SnapFrame prototype build plan

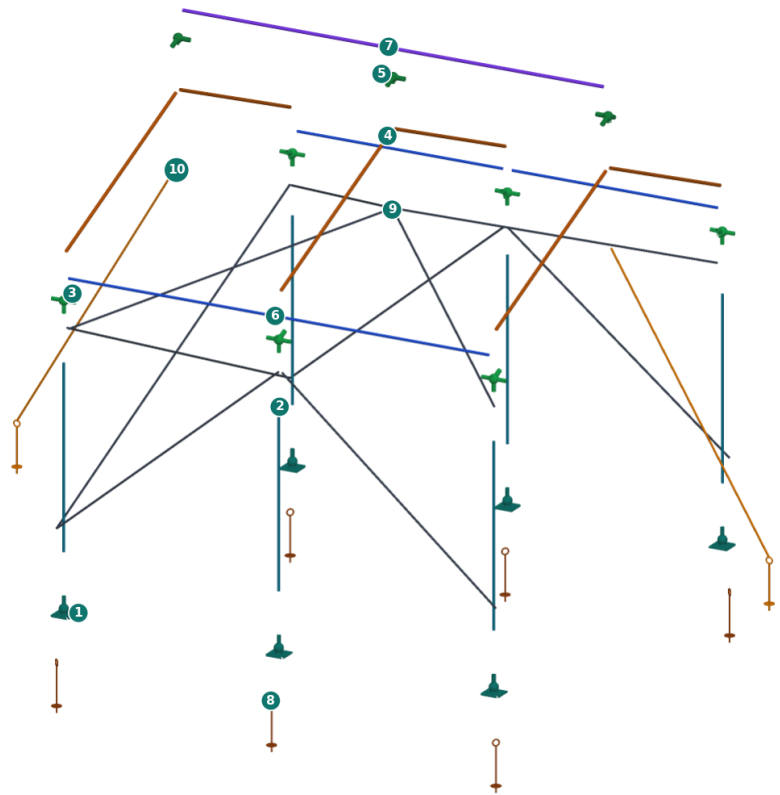
Plan, not yet built. How to build the first proof-of-concept prototype, component by component. Building and testing to it is TRL 4 work. Decisions still to be made are kept in the design decisions register ([docs/06-design-decisions.md](#)), not here.

■ 1. What you are building

SnapFrame size M prototype: every component, pulled apart

Numbered in build order. Cables drawn thicker than they are; snap buttons sit inside the tube ends; tarpaulins not shown

- 1 ■ Foot nodes (6), with hitch pins
- 2 ■ Posts (6), snap buttons inside
- 3 ■ Eave nodes (6) with cable bolts and rings; caps (4)
- 4 ■ Rafters (6)
- 5 ■ Ridge nodes (3), cable bolts and rings
- 6 ■ Eave tubes (4)
- 7 ■ Ridge tubes (2)
- 8 ■ Screw anchors (6 at the feet)
- 9 ■ Brace cables (10) with snap hooks
- 10 ■ Guy lines (2) and guy anchors (2)



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github.com/BoujeeEnjinia1701/snapframe

Figure 1. Every component of the size M frame kit, pulled apart and numbered in build order. The cables are drawn thicker than they are so they show.

The prototype is one size M SnapFrame: a gable-roofed frame 4.0 m square, 1.8 m to the eaves and 2.6 m to the ridge, made of 18 straight steel conduit tubes that click into 15 printed plastic nodes, braced with ten steel cables and held down by eight screw anchors, with two relief tarpaulins tied over it. Figure 1 shows the components in the order you make or fit them. Nine kinds are made: the foot, eave and ridge nodes (printed, then fitted with a steel cable bolt), the posts, rafters, eave tubes and ridge tubes (cut and drilled conduit), and the brace cables (made up from wire rope). The anchors, guy lines, snap buttons, hitch pins, caps and tarpaulins are bought. The work is sawing and drilling thin-wall tube, 3D printing, bolting, and making up wire rope ends with clips. The frame kit's parts cost about \$469 from the bill of materials.

Safety: The finished frame is an emergency shelter frame, not a storm refuge: by calculation it stays elastic only up to gusts of about 19.7 m/s (71 km/h), and less with wind into the open gable. Never put it up, take it down or fit the tarpaulins in strong wind, when the tarpaulin acts as a sail. Cut conduit ends are sharp: deburr every end and wear gloves. Raising a frame takes two people; keep fingers out of the sockets as tubes go in. Check for buried cables and pipes before turning in an anchor.

■ 2. What changed to make it buildable

The concept showed what the frame does; some of its parts could not be made, fixed or put up as drawn. Each change below keeps what the frame does, and all of them are recorded in decision record SNF-DDR-003, open for Amish's review.

Table 1. Changes from the concept.

Component	The concept had	The buildable design has	Why
Tube ends	Spring button 25 mm and hitch pin 50 mm from the tube end	Button 40 mm and hitch pin 15 mm from the end (Figure 6)	The button's spring runs about 45 mm into the tube; the pin went through it
Sockets	Button hole straight through a 5 mm wall; square socket mouths	A 16 mm finger recess round each button hole and a chamfer at each mouth (Figure 5)	A fingertip can press the button to undo the joint, and the button rides in
Eave and ridge nodes	A thin printed tab for the cables on the eave nodes, pointing away from them; nothing on the middle ridge node; cables drawn through the node cores	A stainless bolt through each node with a steel ring on a spacer; every cable and guy clips into a ring (Figures 8, 9 and 13)	A printed tab would bend and break; the bolt takes any pull and the ring swings to it
Foot nodes	Cables drawn to the node centres; the anchor eye 32 mm above the plate; one anchor slot, straight out	The lower cable ends clip to the anchor eye; the eye is turned down onto the plate; two slots, at 45° either way (Figure 3)	The cable pull goes straight into the anchor, the plate is held from the first turn, and the gable cable passes clear of the foot
Erection order	Anchors first, frames dropped onto the anchored feet, then the tubes between them	Frames stand unanchored; each next frame slides 65 mm on its feet onto the tube ends; anchors go in last (steps 4 to 8)	A tube cannot be fitted between two fixed sockets that are closer together than the tube is long
Tube drilling	Not stated	The two button holes of every tube on opposite sides, on one marked line (Figure 4)	One drilling rule; every tube of a kind fits any position

■ 3. Making the components

Make and check each component before the assembly step that needs it. Sizes are in millimetres. "Inboard" means toward the middle of the shelter and "outboard" away from it; the "rear" gable is the closed one and the "front" gable is the open doorway. Workshop tolerance is 1 mm on tube lengths and 0.5 mm on hole positions unless a step says otherwise; drawings do not carry tolerances before TRL 4.

3.1 Foot nodes (make 6)

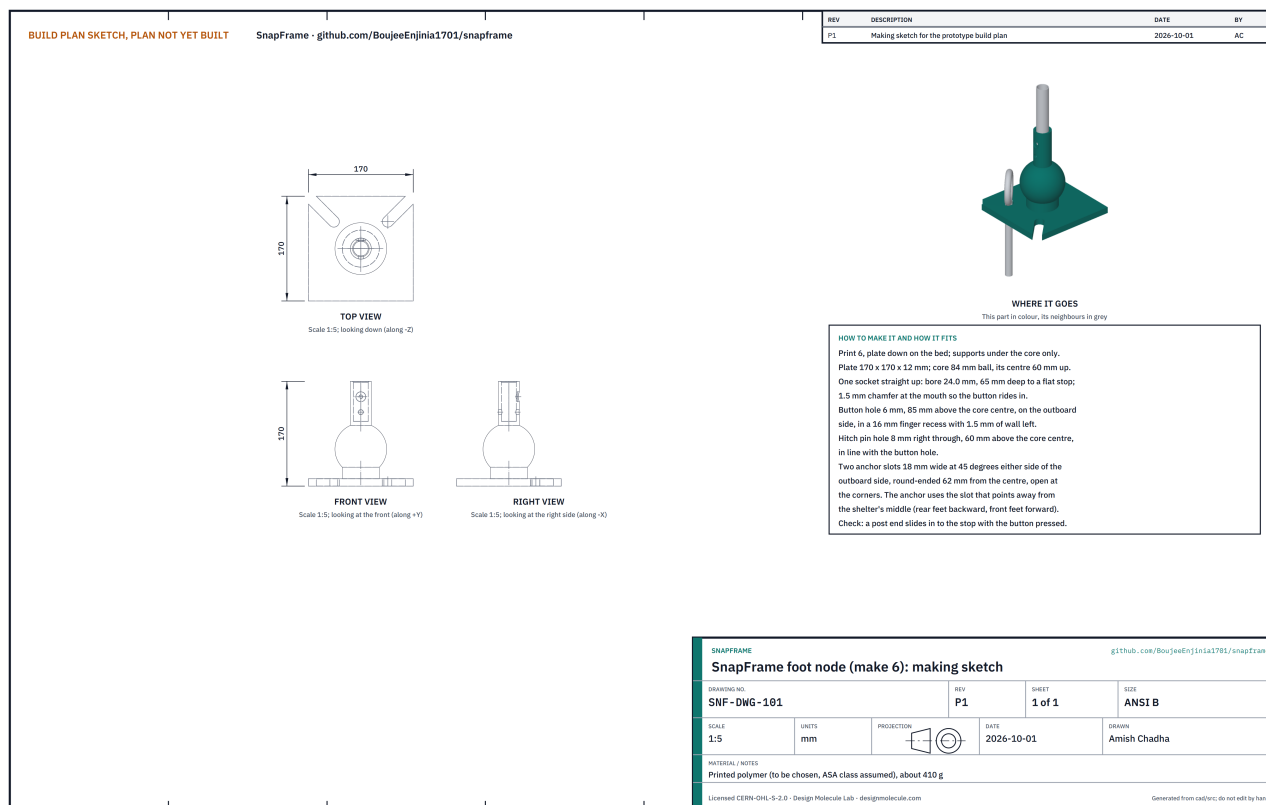


Figure 2. Foot node making sketch (SNF-DWG-101).

What it is and what it is made from. The node the bottom of each post stands in: an 84 mm ball with one socket pointing straight up, on a square plate that lies on the ground and is held down by a screw anchor. Printed in a glass- or carbon-filled nylon of the PA12 class (decided 2026-10-02, SNF-DEC-001), about 410 g each in ASA (to be re-estimated for the nylon).

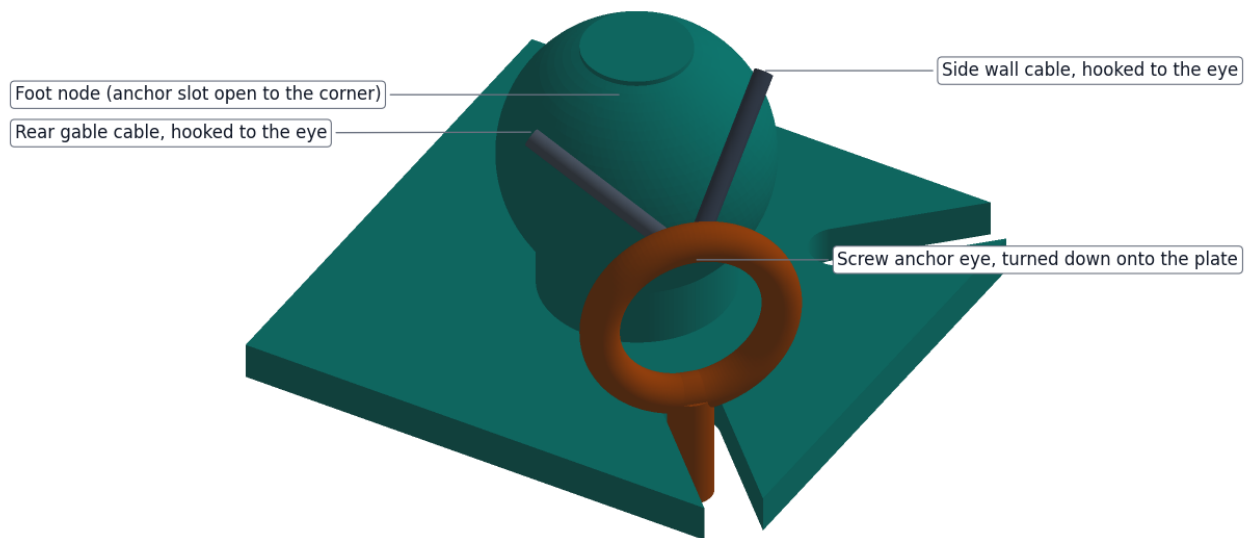
How to make it.

1. Print six, plate down on the bed, with supports under the ball only. Use the print settings that give the calculation note's assumption of about 55 % solid: four or more walls, and 40 % infill or more.
2. Let each print cool on the bed so the plate stays flat. Remove the supports and clean the socket bore.
3. Check the socket bore with a short offcut of 3/4 in conduit: it must slide in to the flat stop at the bottom of the bore without force. Ream lightly if the print is tight.
4. Check the button hole and its finger recess on the outboard side of the socket, and the hitch pin hole through the socket in line with it; clear any stringing with a 6 mm and an 8 mm drill turned by hand.
5. Check both anchor slots: 18 mm wide, open to the plate corners at 45° either side of the outboard side, round-ended 62 mm from the centre.

How it fits the parts next to it. A post goes into the socket until its end sits on the flat stop; its button clicks into the hole and a hitch pin goes through (Figure 6, in section 3.2). The plate lies flat on the ground on its mark. A screw anchor goes down through one slot, the one that points away from the middle of the shelter (backward at the rear feet, forward at the front feet, either at the middle feet), and its eye is turned down until it sits on the plate across the slot:

Joint 2: anchor on the foot plate, rear corner

Seen from outside and above. The eye sits down on the plate across the slot; both cables clip to the eye



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Figure 3. The anchor eye sits down on the plate across the slot; at a corner, the gable cable and the side wall cable clip to the same eye.

Check before moving on. The plate lies flat on a flat floor without rocking; a post offset clicks in and the hitch pin goes through.

3.2 Posts (make 6)

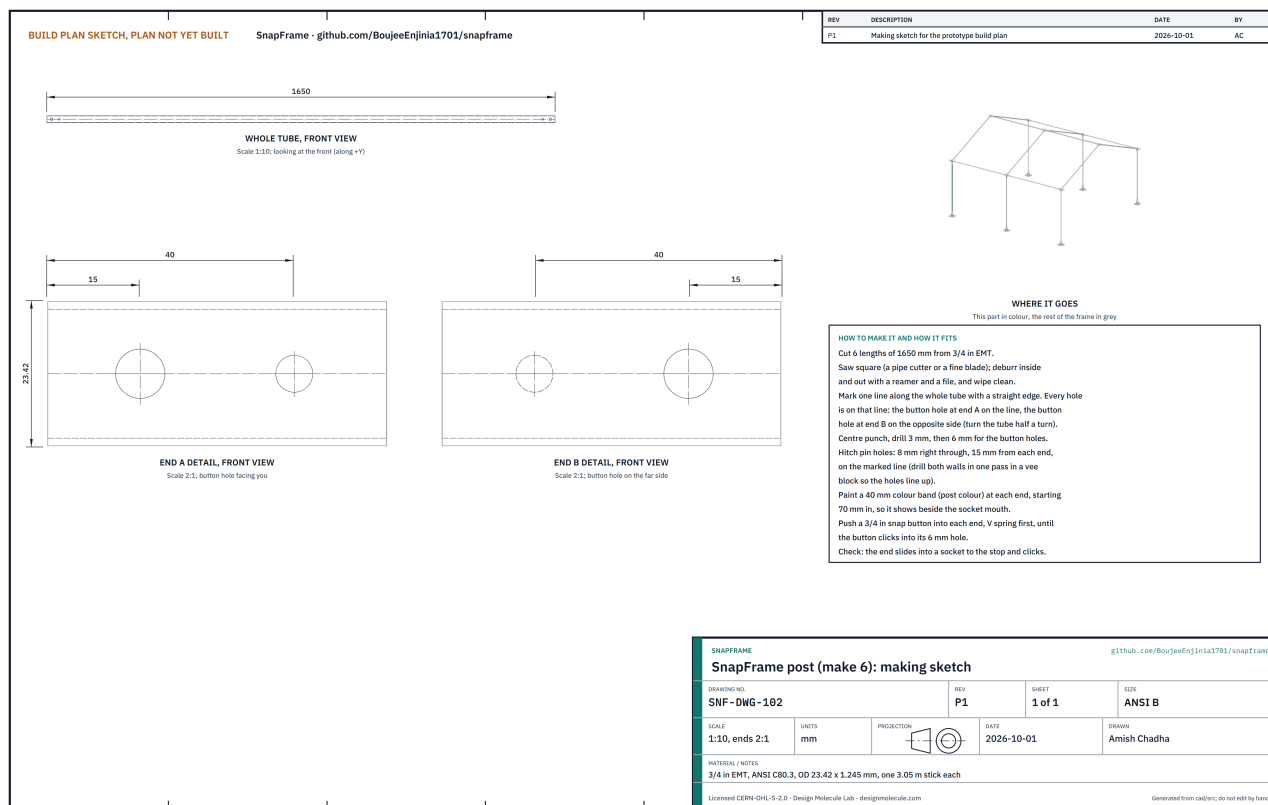


Figure 4. Post making sketch (SNF-DWG-102): the whole tube, and both ends drawn twice full size, with the button hole facing you at one end and on the far side at the other.

What it is and what it is made from. The upright at each corner and in the middle of each side wall. 3/4 in EMT conduit (23.42 mm outside, 1.245 mm wall), cut to 1,650 mm, one from each 3.05 m stick.

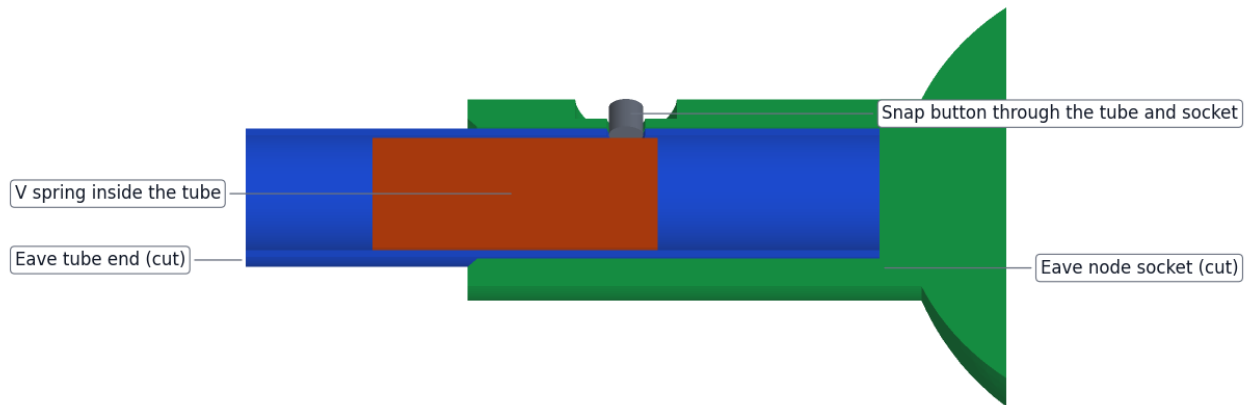
How to make it.

1. Cut six lengths of 1,650 mm, square, with a pipe cutter or a fine-tooth blade. Ream the inside and file the outside edge of each end smooth, and wipe off the swarf.
2. Lay each tube on a straight edge and draw one line along its whole length.
3. Mark the holes on the line: a hitch pin hole 15 mm from each end, and a button hole 40 mm from end A. Turn the tube half a turn and mark the button hole 40 mm from end B, on the opposite side.
4. Centre punch every mark. In a vee block, drill the hitch pin holes 3 mm, then 8 mm, straight through both walls in one pass so the two sides line up. Drill the button holes 3 mm, then 6 mm, through one wall only.
5. Deburr every hole inside and out.
6. Paint a 40 mm colour band in the post colour round each end, starting 70 mm from the end, so it shows beside the socket mouth.
7. Squeeze a 3/4 in snap button, spring first, into each end and slide it in until the button clicks out through its hole.

How it fits the parts next to it.

Joint 6: tube end in a socket (cut open), middle eave node

The button stands 2 mm proud in a 16 mm finger recess: press it with a fingertip and pull the tube



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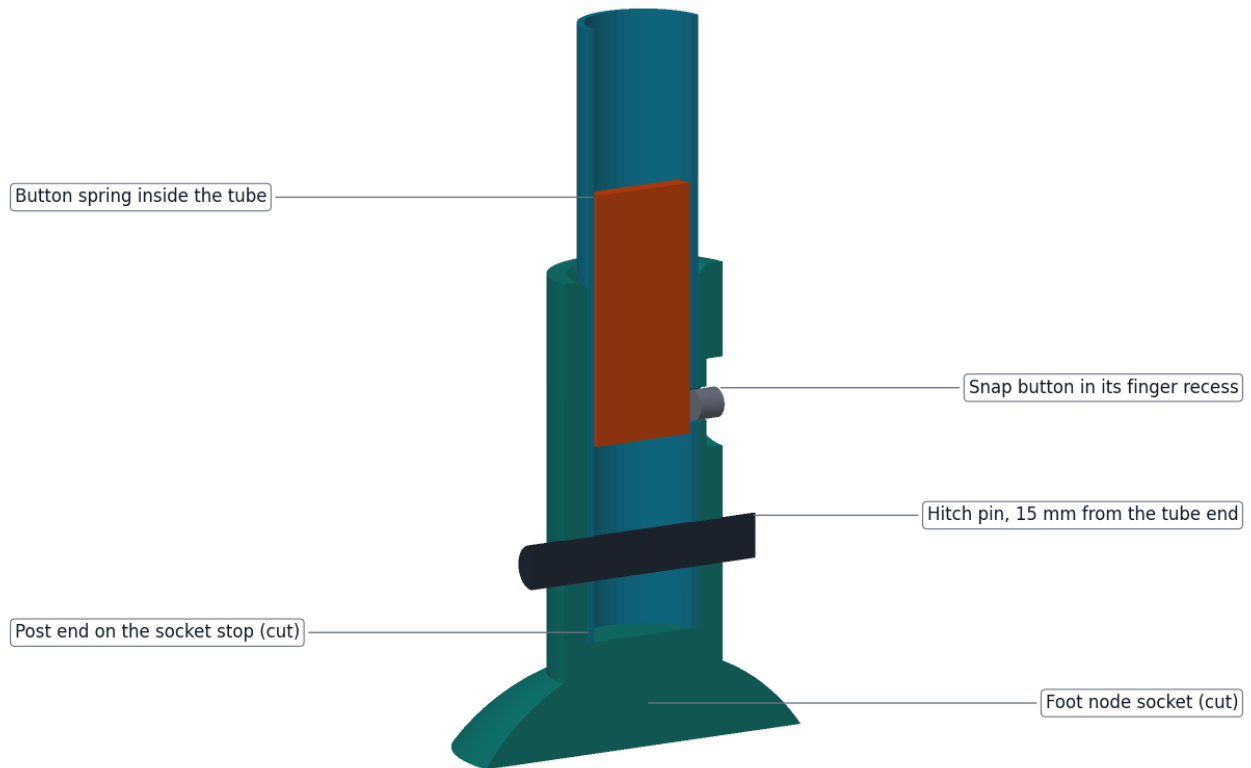
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Figure 5. Every tube end fits its socket the same way (an eave tube shown): the end sits on the flat stop, the button stands about 2 mm proud in the finger recess, and the spring lies inside the tube.

Each end slides into its socket with the button pressed, until the end sits on the flat stop 65 mm in, and the button clicks out into the socket's hole. The bore is 0.6 mm larger than the tube, so the tube slides by hand. To undo a joint, press the button down into its recess with a fingertip and pull the tube out. Both ends of a post are pinned: a hitch pin goes through the socket and the tube 15 mm from the tube end, clear of the button's spring:

Joint 1: post in the foot node (cut open through the post)

Seen from inboard. The tube end sits on the flat stop; pin 15 mm and button 40 mm from the end; the spring clears the pin



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Figure 6. The bottom of a post in the foot node, cut open through the post. The eave end is pinned the same way.

Check before moving on. Lengths within 1 mm; each end clicks into a test socket and comes out by hand; both hitch pin holes take an 8 mm pin straight through.

3.3 Eave nodes and their cable bolts (make 6)

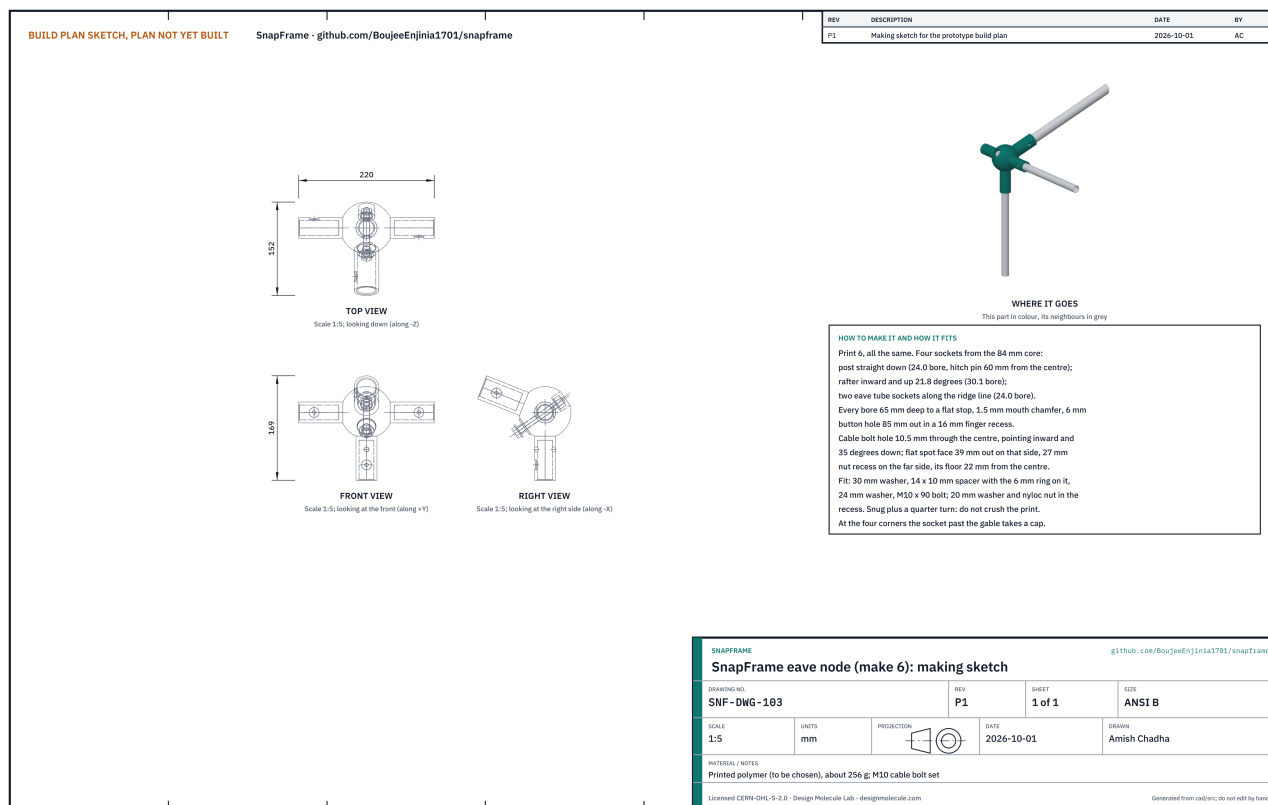


Figure 7. Eave node making sketch (SNF-DWG-103), with its cable bolt set.

What it is and what it is made from. The node at the top of each post, where the post, the rafter and the eave tubes meet. One design serves all six places: at the four corners the socket that points past the gable is left empty and capped. Printed in the same polymer, about 256 g, with a bought stainless cable bolt set through it.

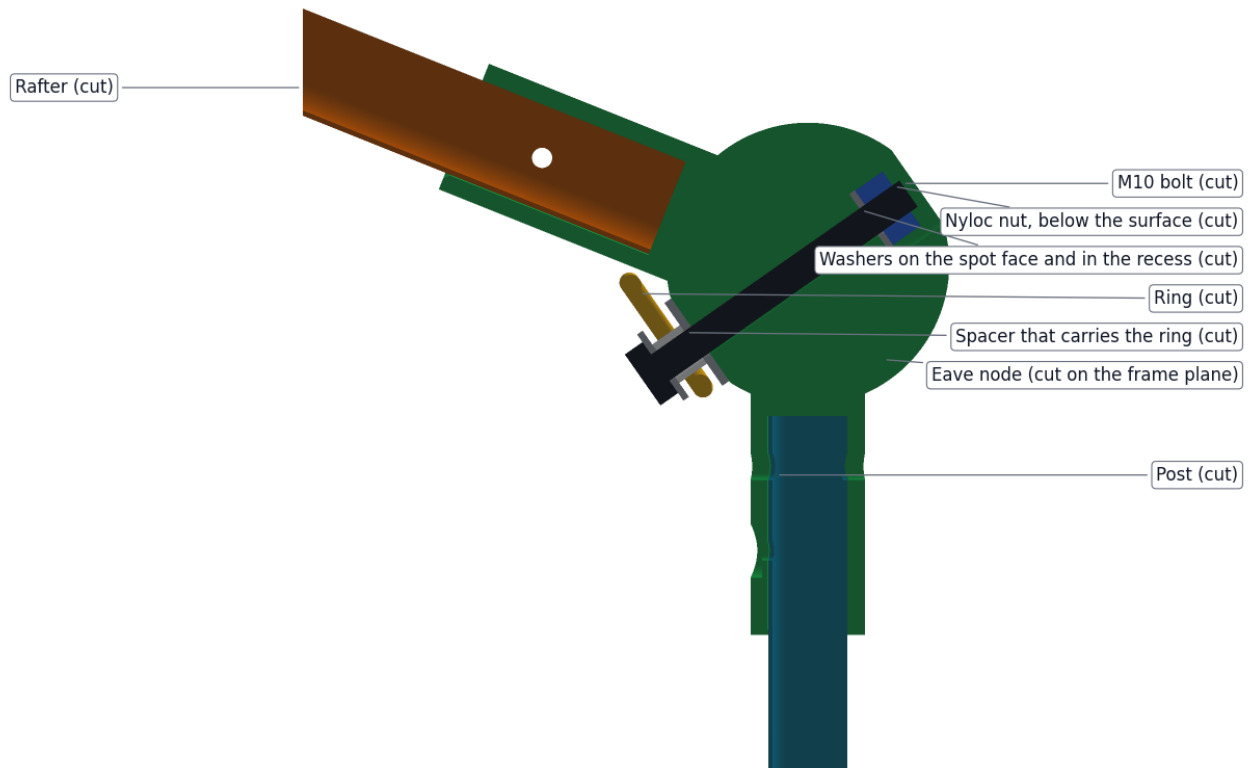
How to make it.

1. Print six, with supports. The node measures 220 x 152 x 169 mm, so the printer bed must be at least 230 mm across.
2. Clean and check each socket as for the foot node: the post socket (straight down) and the two eave tube sockets (level, along the ridge line) take 3/4 in conduit; the rafter socket (inboard, rising 21.8°) takes 1 in conduit. The post socket has a hitch pin hole.
3. Check the cable bolt hole: 10.5 mm, straight through the ball, pointing inboard and 35° down, with a flat spot face on that side and a 27 mm recess on the far side.
4. Fit the cable bolt set from the spot face side: the 30 mm washer on the spot face, the 14 mm spacer with the 6 mm steel ring threaded on it, the 24 mm washer, then the M10 x 90 bolt through all of them and the ball. In the recess fit the 20 mm washer and the M10 nyloc nut.
5. Tighten the nut with a deep socket until snug, then a further quarter turn. Do not crush the print: the bolt clamps the washer and spacer, not the ring.

How it fits the parts next to it.

Joint 4: cable bolt through the eave node (cut open)

Seen along the ridge. Washer on a flat spot face; spacer carries the ring; nut below the surface in its recess



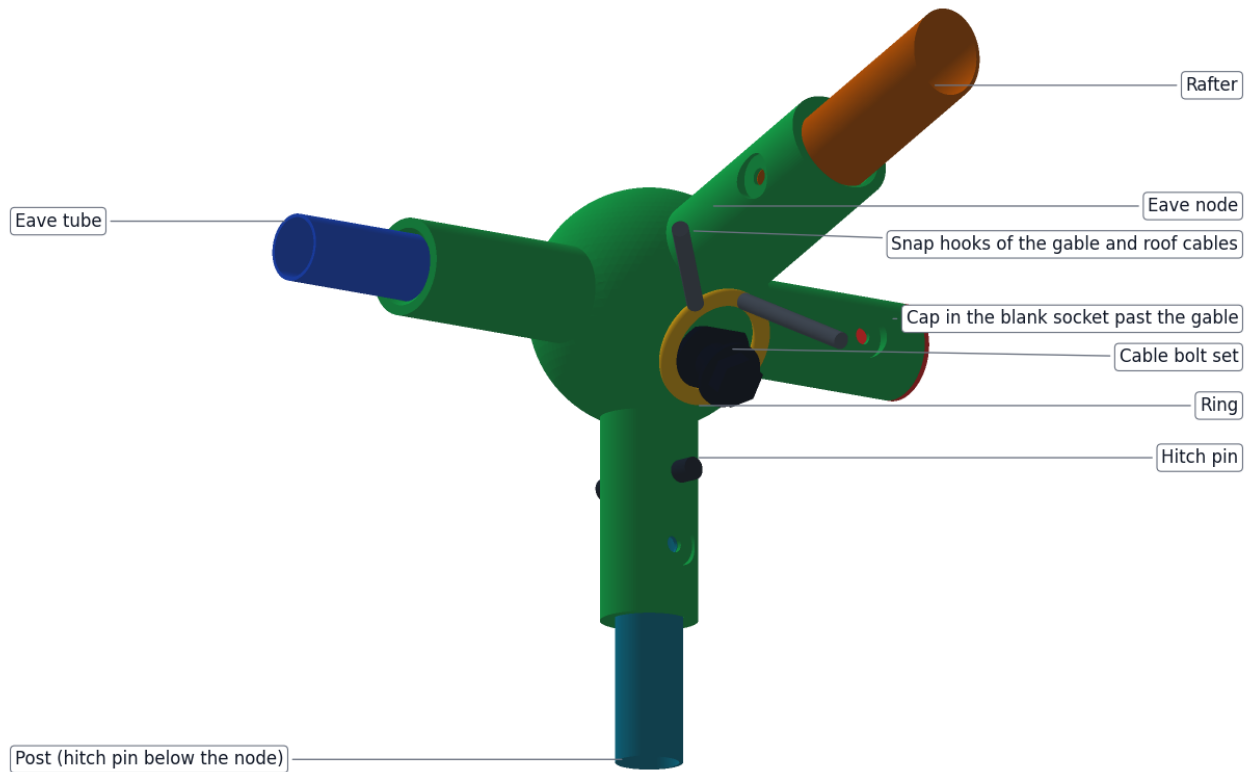
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Figure 8. The cable bolt through the eave node, cut open along the frame. The washer sits on a flat spot face, the spacer carries the ring, and the nut sits below the surface in its recess.

Joint 3: rear corner eave node, from inside the shelter

Four sockets; the one past the gable is capped. Cables clip to the ring on the cable bolt, never to the print



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Figure 9. A rear corner eave node: the post below (pinned), the rafter rising inboard, the eave tube along the side wall, the cap in the socket past the gable, and two cables clipped into the ring.

The post, rafter and eave tubes click in as in Figure 5. The ring swings freely on the spacer and lines up with the cables clipped into it; the cables never touch the printed node.

Check before moving on. The ring swings all round the spacer by hand; nothing stands proud of the far side of the ball; every socket takes its tube to the stop.

3.4 Rafters (make 6)

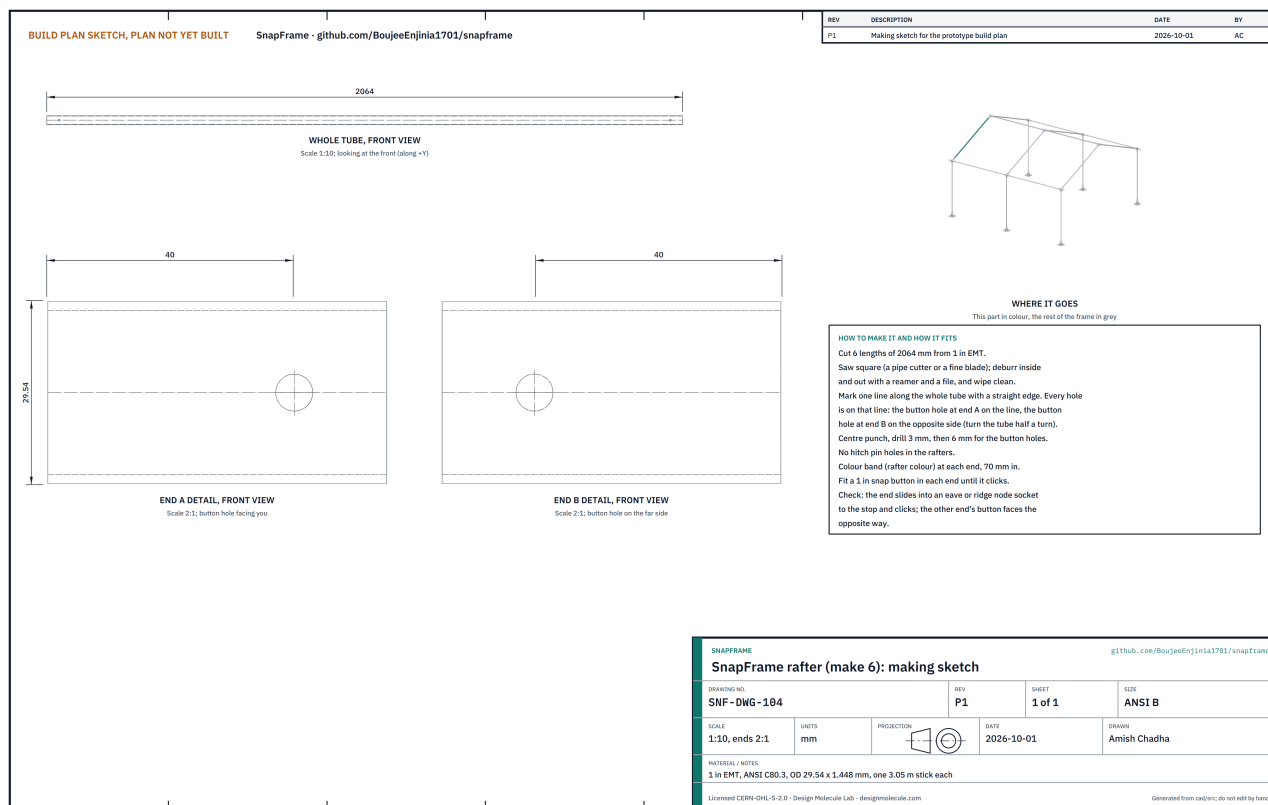


Figure 10. Rafter making sketch (SNF-DWG-104).

What it is and what it is made from. The sloping tube from each eave node up to the ridge node. 1 in EMT conduit (29.54 mm outside, 1.448 mm wall), cut to 2,064 mm, one from each 3.05 m stick.

How to make it.

1. Cut six lengths of 2,064 mm; ream and file both ends.
2. Draw a line along each tube; mark a button hole 40 mm from end A on the line and 40 mm from end B on the opposite side. There are no hitch pin holes in the rafters.
3. Drill 3 mm, then 6 mm, through one wall; deburr.
4. Paint the rafter colour band at each end, 70 mm in.
5. Fit a 1 in snap button in each end until it clicks.

How it fits the parts next to it. The low end goes into the eave node's rafter socket and the high end into a ridge node socket, as in Figure 5. With the button at the low end facing one way, the button at the high end faces the other: that is how the sockets are made.

Check before moving on. Length within 1 mm; both ends click into 1 in sockets.

3.5 Ridge nodes, end (make 2)

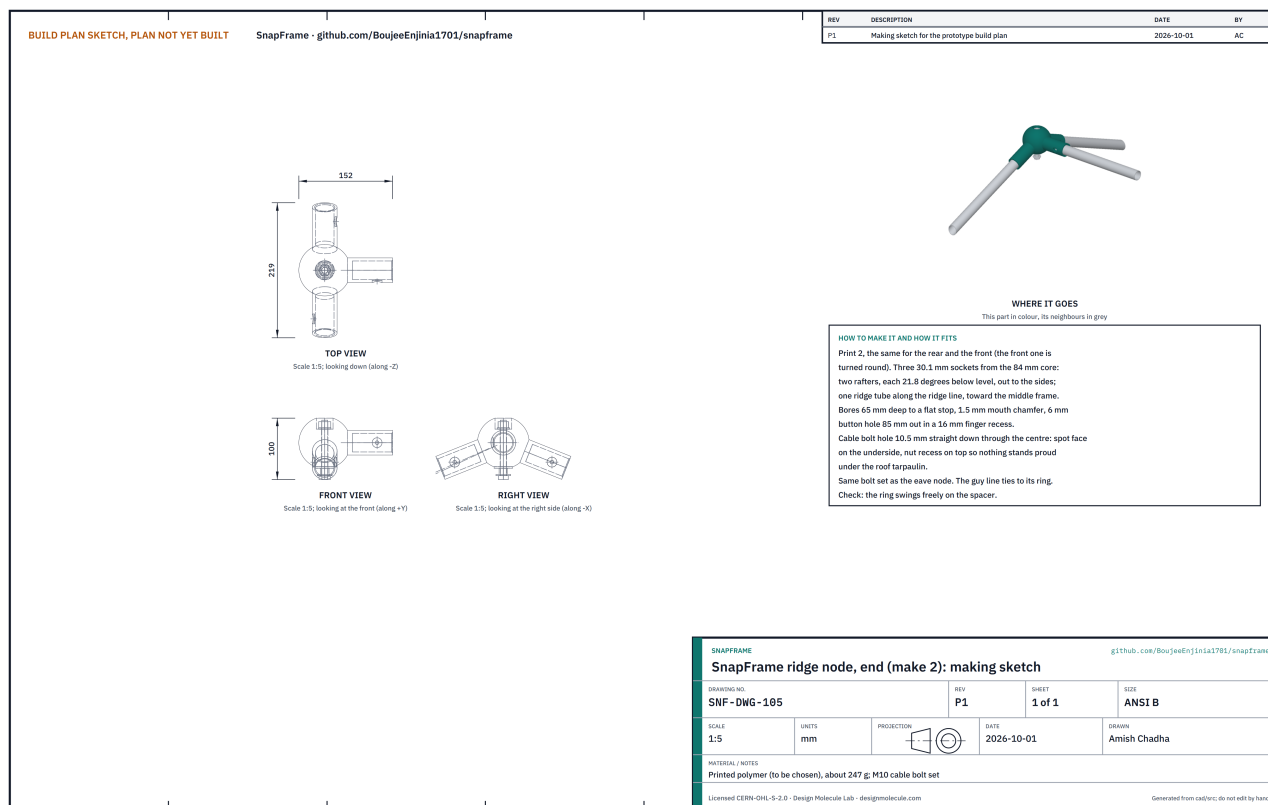


Figure 11. End ridge node making sketch (SNF-DWG-105).

What it is and what it is made from. The node at the top of the rear frame and of the front frame: two rafter sockets sloping down to the sides at 21.8° and one ridge tube socket pointing along the ridge toward the middle frame, all for 1 in conduit. The rear and front nodes are the same; the front one is turned round. Printed, about 247 g, with a cable bolt set pointing straight down.

How to make it.

1. Print two, with supports; clean and check each socket with a 1 in offcut.
2. Fit the cable bolt set as for the eave node (section 3.3, steps 4 and 5), from below: the spot face is on the underside and the nut recess on top, so nothing stands proud under the roof tarpaulin.

How it fits the parts next to it. The two rafters and the ridge tube click in as in Figure 5. The guy line for that gable ties to the ring.

Check before moving on. The ring swings freely; the top of the node is smooth.

3.6 Ridge node, middle (make 1)

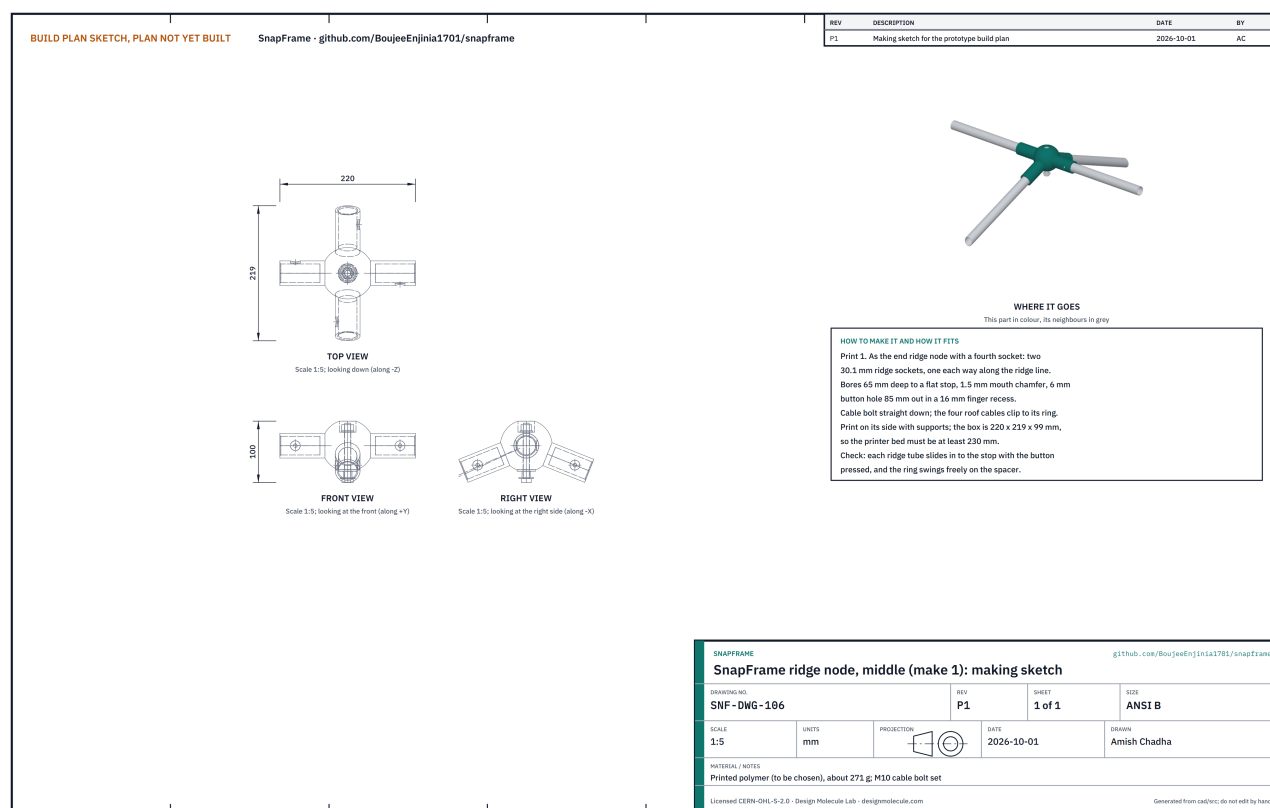


Figure 12. Middle ridge node making sketch (SNF-DWG-106).

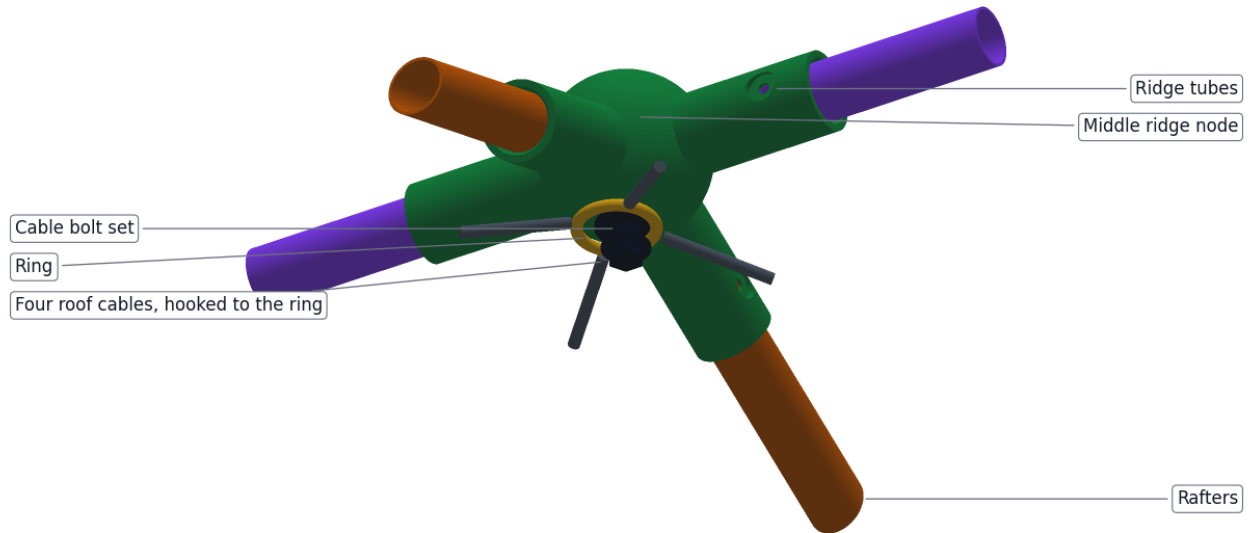
What it is and what it is made from. As the end ridge node, with a fourth socket so that a ridge tube goes each way along the ridge. Printed, about 271 g (220 x 219 x 99 mm), with a cable bolt set pointing straight down.

How to make it. Print one and fit its cable bolt set as in section 3.5.

How it fits the parts next to it.

Joint 5: middle ridge node from below

All four roof cables clip into the one ring; the ring swings to line up with the pull



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Figure 13. All four roof cables clip into the one ring below the middle ridge node.

Check before moving on. Each ridge socket takes a 1 in offcut to the stop; the ring swings freely.

3.7 Eave tubes (make 4)

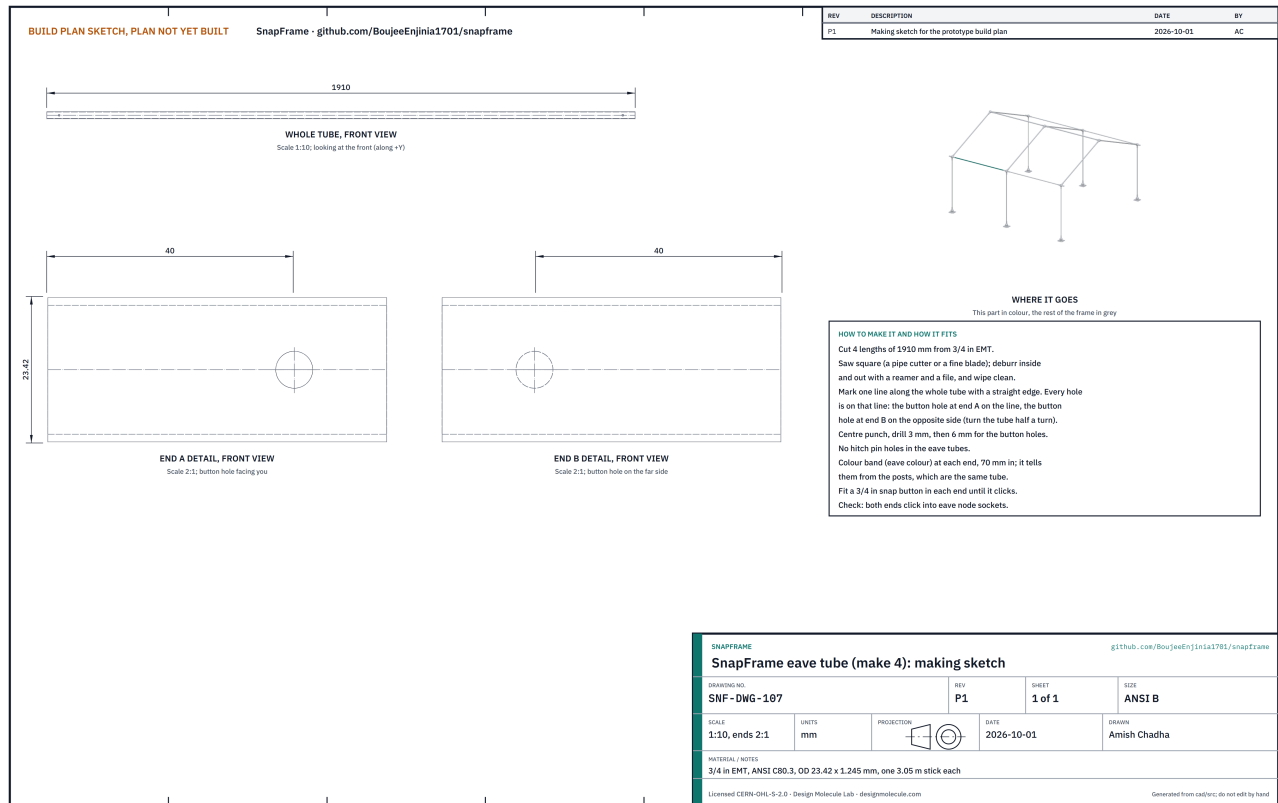


Figure 14. Eave tube making sketch (SNF-DWG-107).

What it is and what it is made from. The level tube along the top of each side wall, from eave node to eave node. 3/4 in EMT, cut to 1,910 mm.

How to make it. As the rafter (section 3.4) with 3/4 in conduit and length 1,910 mm: button holes 40 mm from each end on opposite sides, no hitch pin holes, the eave colour band at each end (it tells them from the posts, which are the same conduit), and a 3/4 in snap button in each end.

How it fits the parts next to it. Each end clicks into an eave node's level socket (Figure 5).

Check before moving on. Length within 1 mm; both ends click in.

3.8 Ridge tubes (make 2)

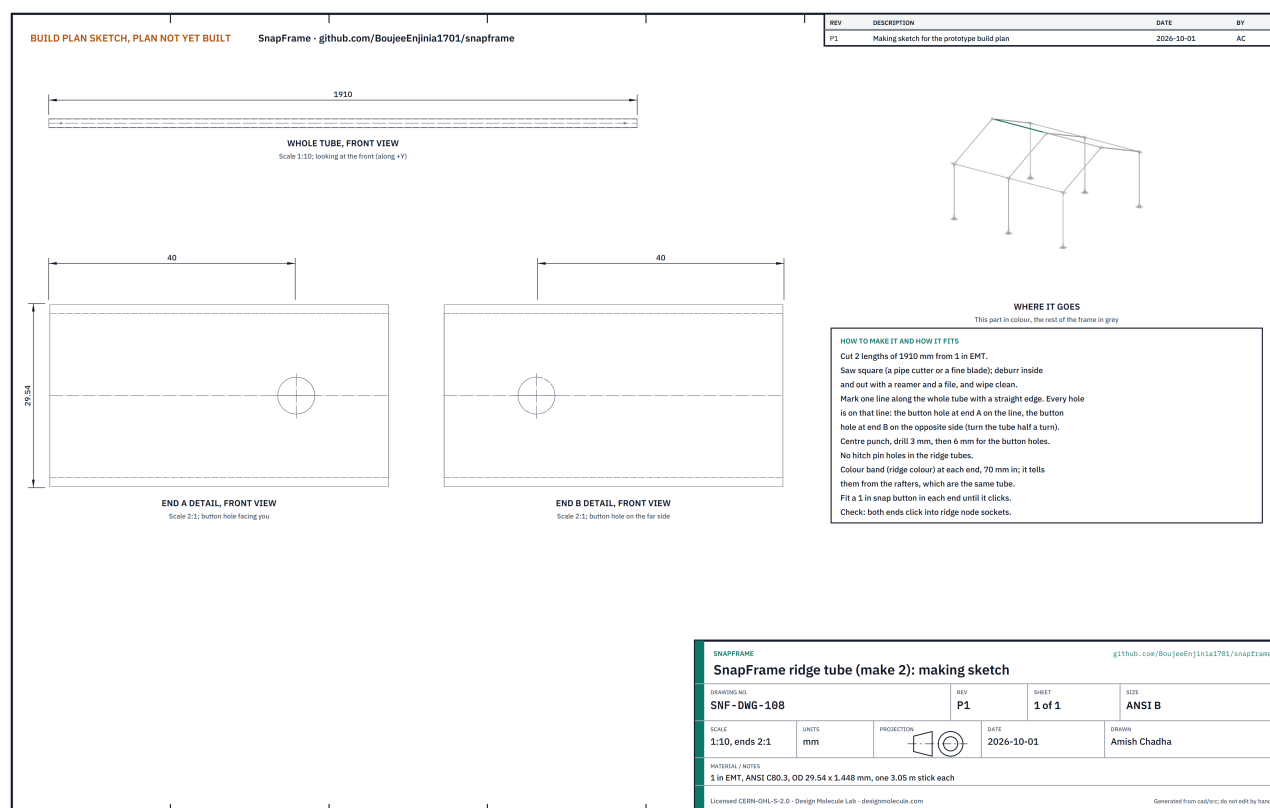


Figure 15. Ridge tube making sketch (SNF-DWG-108).

What it is and what it is made from. The level tube along the ridge between ridge nodes. 1 in EMT, cut to 1,910 mm.

How to make it. As the rafter (section 3.4), length 1,910 mm, with the ridge colour band at each end (it tells them from the rafters, which are the same conduit).

How it fits the parts next to it. Each end clicks into a ridge node's ridge socket (Figure 5).

Check before moving on. Length within 1 mm; both ends click in.

3.9 Brace cables (make 10)

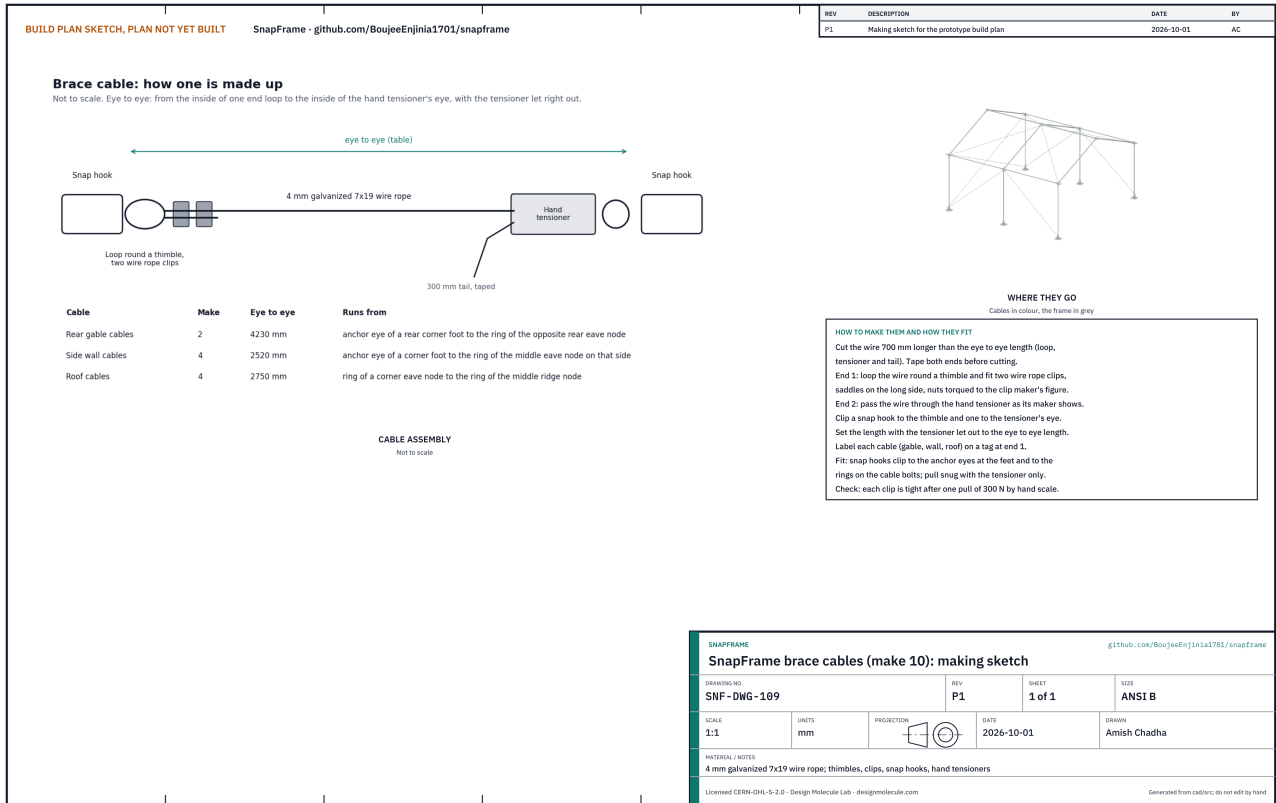
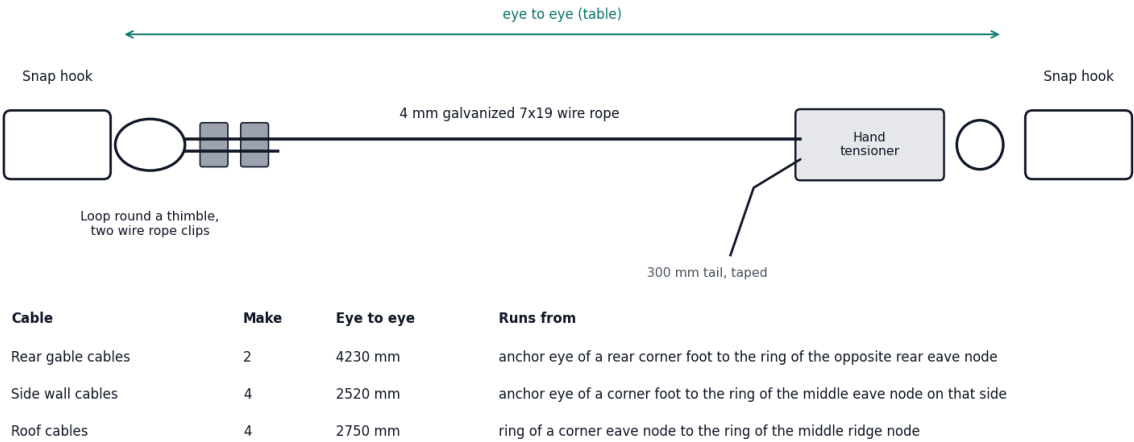


Figure 16. Brace cable making sketch (SNF-DWG-109).

Brace cable: how one is made up

Not to scale. Eye to eye: from the inside of one end loop to the inside of the hand tensioner's eye, with the tensioner let right out.



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Figure 17. One brace cable, with the three lengths needed.

What it is and what it is made from. A steel wire rope with a snap hook at each end and a hand tensioner at one end, that braces the frame against wind. Ten are needed: two for the rear gable (4,230 mm eye to eye), four for the side walls (2,520 mm) and four for the roof (2,750 mm). 4 mm galvanized 7x19 wire rope, thimbles, wire rope clips, hand tensioners and snap hooks.

How to make it.

1. Tape the wire where you will cut it and cut each length 700 mm longer than its eye-to-eye length, to allow for the loop, the tensioner and a tail.
2. At one end, bend the wire round a thimble and fit two wire rope clips, the saddles on the long side of the wire. Tighten the clip nuts to the clip maker's figure.
3. Thread the other end through the hand tensioner as its maker shows. Set the tensioner let right out and adjust the wire so the cable measures its eye-to-eye length. Tape the tail.
4. Clip a snap hook to the thimble and one to the tensioner's eye.
5. Tag each cable "gable", "wall" or "roof" at the thimble end.

How it fits the parts next to it. The gable and side wall cables run from an anchor eye at a corner foot (Figure 3) up to the ring of an eave node; the roof cables run from the ring of a corner eave node to the ring of the middle ridge node (Figure 13). They are pulled snug with the tensioner only.

Check before moving on. Each cable within 20 mm of its eye-to-eye length; the clips do not slip when the cable is pulled to 300 N on a hand scale.

3.10 Bought components

Buy to specification, not brand. Line numbers are those of the bill of materials.

- **Screw anchors (line 9).** Eight galvanized screw anchors about 380 mm long with a 90 mm helix and an eye of about 12 mm wire and 56 mm across, so the eye straddles the 18 mm slot. Six go at the feet, two for the guys.
- **Guy lines (line 10).** Two 3.5 m lengths of 6 mm polyester rope, each with a slide tensioner and reflective tape.
- **Tarpaulins (line 11).** Two 4 x 6 m reinforced woven polyethylene tarpaulins with eyelets, relief agency class, from agency stock.
- **Snap buttons, hitch pins and caps (line 12).** 36 stainless spring snap buttons with a 6 mm button: 20 for 3/4 in conduit and 16 for 1 in conduit, each with its spring reaching no more than 45 mm beyond the button and the button standing at least 3.5 mm above the tube. Twelve 8 mm hitch pins on lanyards. Four push-in caps for the 24.0 mm corner sockets.
- **Bundle straps and bag (line 13).** Two 25 mm cam straps and a duffel bag of about 0.08 m³.
- **Cable bolt sets (line 14).** Nine sets, stainless A4-70: an M10 x 90 hex bolt with 26 mm of thread, an M10 nyloc nut, washers of 30, 24 and 20 mm outside diameter, a spacer 14 mm outside and 10 mm long, and a welded ring of 6 mm wire and 32 mm bore.
- **Snap hooks (line 8).** Galvanized or stainless, gate opening 14 mm or more so they pass over the anchor eye wire; two per cable.

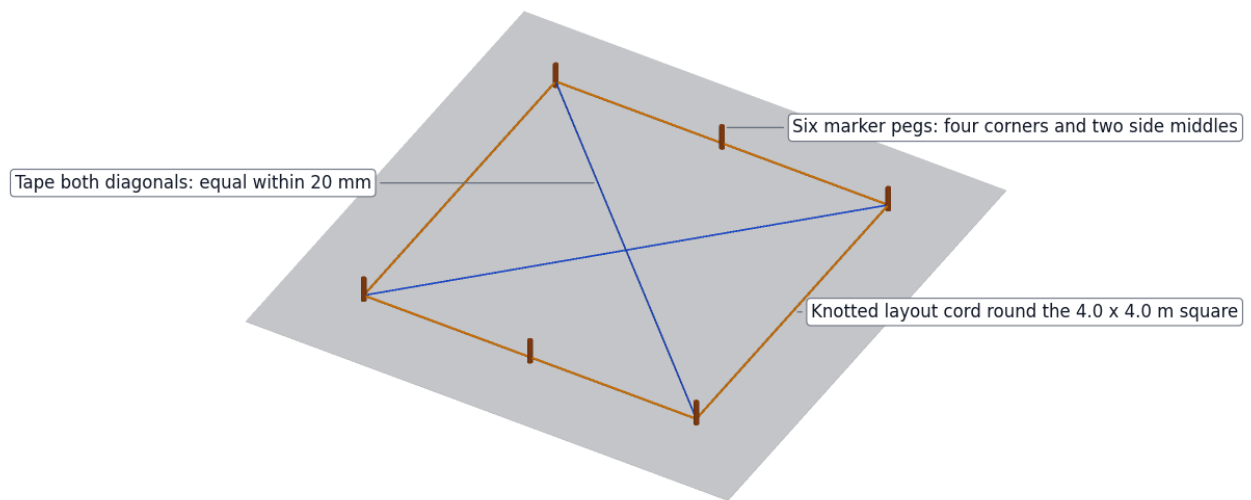
■ 4. Putting it together

In each picture the parts already fitted are grey and the parts being fitted are in colour, with an arrow showing the way they go in. Work with two people, in light wind only.

Step 1: mark out the floor

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Peg the corners and side middles of a 4.0 x 4.0 m square; move pegs until the two diagonals are equal



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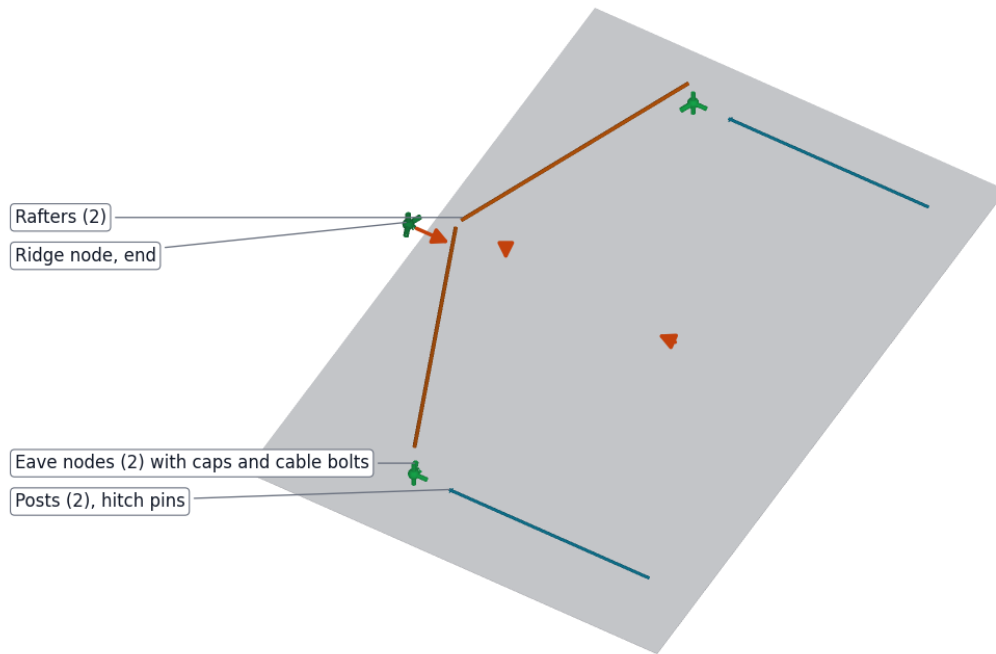
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Peg the four corners and the two middles of the side walls of a 4.0 x 4.0 m square with the knotted layout cord. Measure both diagonals; move pegs until they are equal.

Step 2: build the rear frame flat on the ground

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Posts into the eave nodes with hitch pins; rafters into the eave nodes and the ridge node until every button clicks



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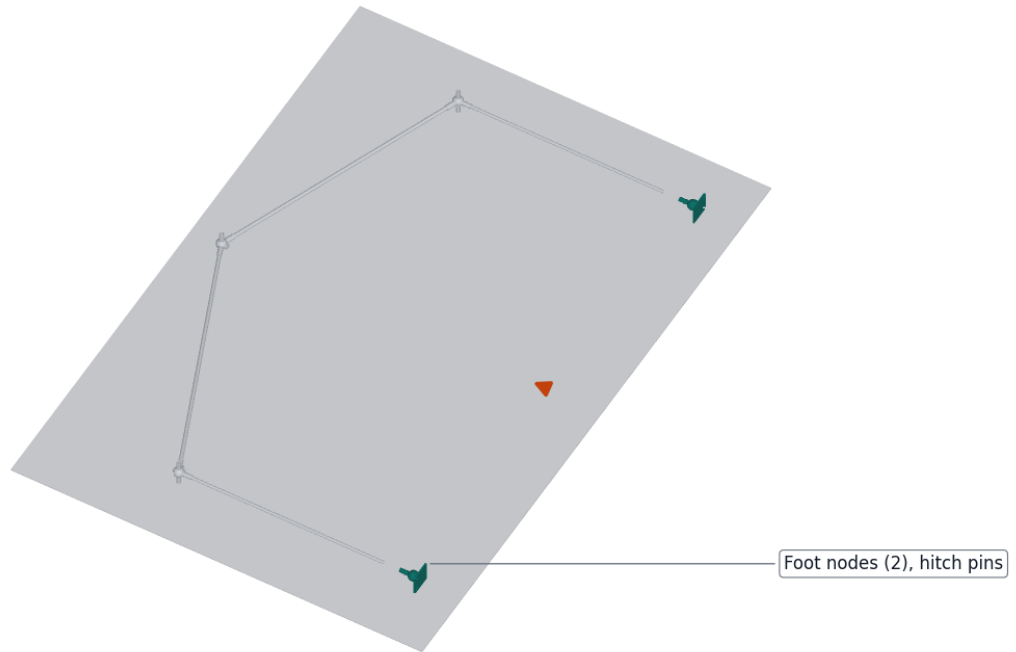
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Lay the two eave nodes, two posts, two rafters and one end ridge node on the ground behind the rear pegs, as the frame will stand. Push each post into the post socket of an eave node until it clicks and put a hitch pin through. Push the rafters into the eave nodes and then into the ridge node until every button clicks. Push a cap into the socket that points away from the shelter on each eave node.

Step 3: foot nodes onto the post ends

Step 3: foot nodes onto the post ends

Each foot over its post end, button clicked, hitch pin through; slots turned to point out and back



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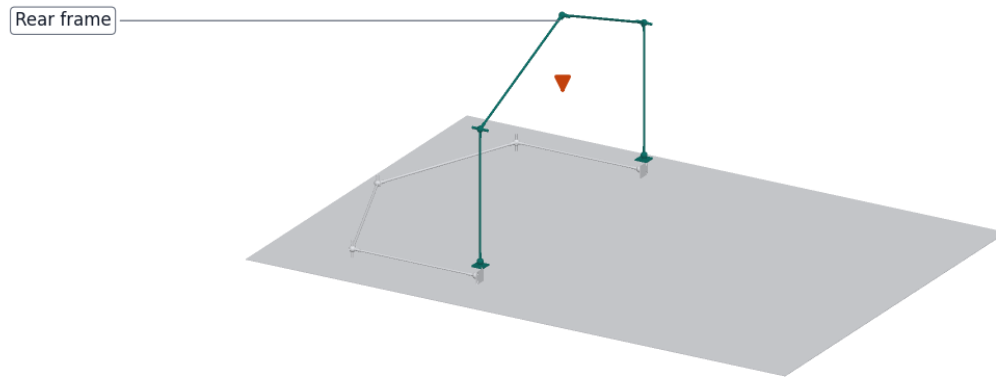
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Push a foot node onto each post end until it clicks and put a hitch pin through. Turn each foot so its slots point outboard. Build the middle frame (no caps) and the front frame (with caps) the same way.

Step 4: stand the rear frame on its marks

Step 4: stand the rear frame on its marks

Two people walk it up from the ridge end (where it lay flat, faint); feet on the rear pegs. One holds it upright until step 6



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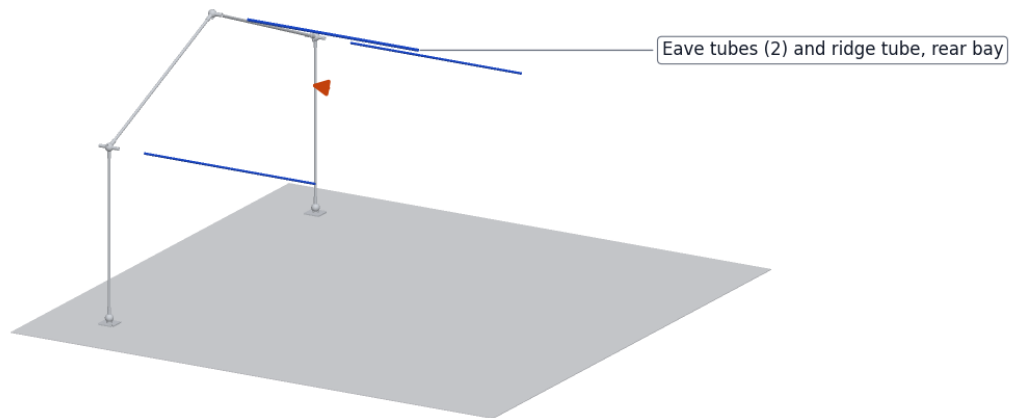
github.com/BoujeeEnjinia1701/snapframe

One person lifts the ridge end and walks toward the feet while the other holds the feet down; stand the frame upright with its feet on the rear pegs (pull the pegs). One person holds it upright until step 6.

Step 5: eave tubes and ridge tube into the rear frame

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Push each straight into its socket until it clicks; a helper holds the far ends level (a step for the ridge)



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

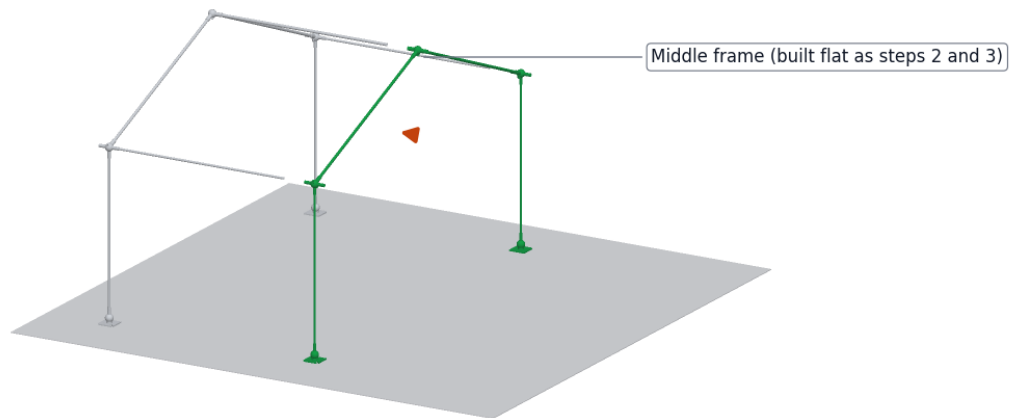
github.com/BoujeeEnjinia1701/snapframe

Push the two eave tubes and the ridge tube into the rear frame's sockets that point along the shelter, until each clicks. Hold the far ends level; use a step to reach the ridge tube.

Step 6: slide the middle frame onto the tube ends

Step 6: slide the middle frame onto the tube ends

Stand it 70 mm short of its pegs, line up the three sockets, slide it back 65 mm on its feet until all three click



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

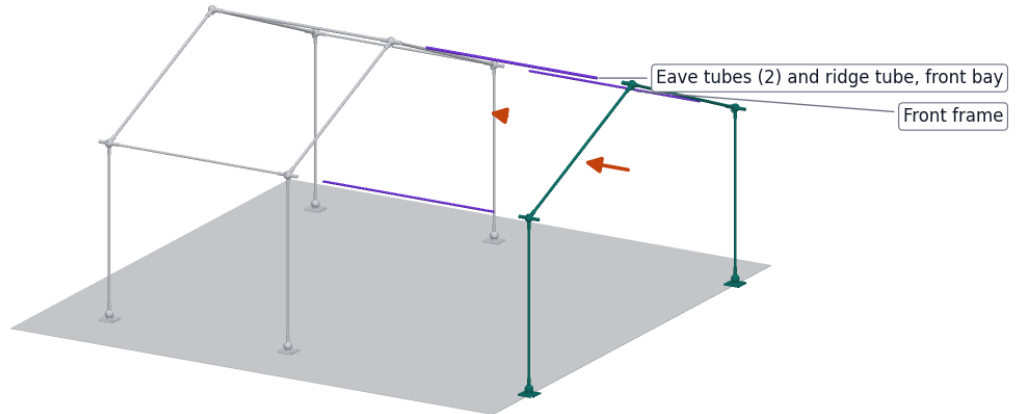
github.com/BoujeeEnjinia1701/snapframe

Stand the middle frame about 70 mm beyond its pegs. Line up its three sockets with the three tube ends, then slide the whole frame 65 mm on its feet toward the rear frame until all three buttons click. **Hold point:** all three joints clicked before anyone lets go.

Step 7: front bay: tubes, then the front frame

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Tubes into the middle frame; slide the front frame on as in step 6; caps go in the two front corner sockets



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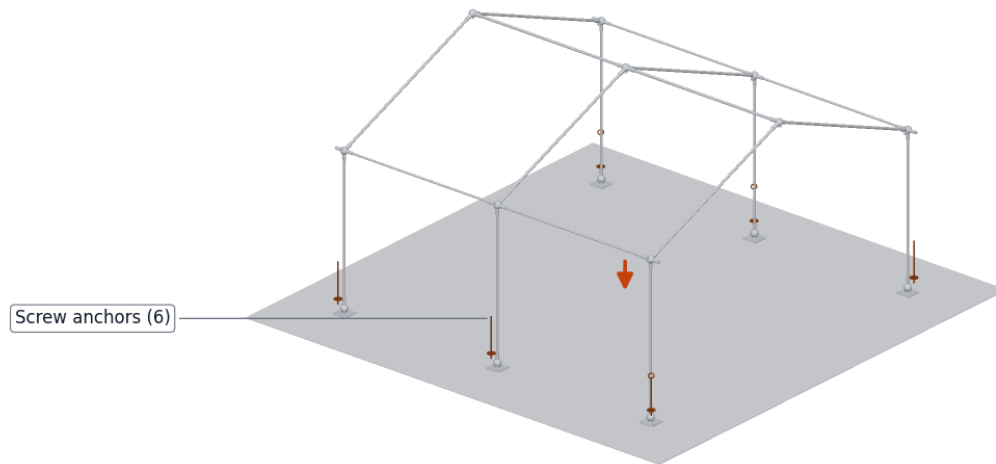
github.com/BoujeeEnjinia1701/snapframe

Push the front bay's two eave tubes and ridge tube into the middle frame, then slide the front frame onto them as in step 6.

Step 8: square the frame and turn in the anchors

Step 8: square the frame and turn in the anchors

Diagonals equal; each anchor down through its slot, turned by a spare tube through the eye until the eye sits on the plate



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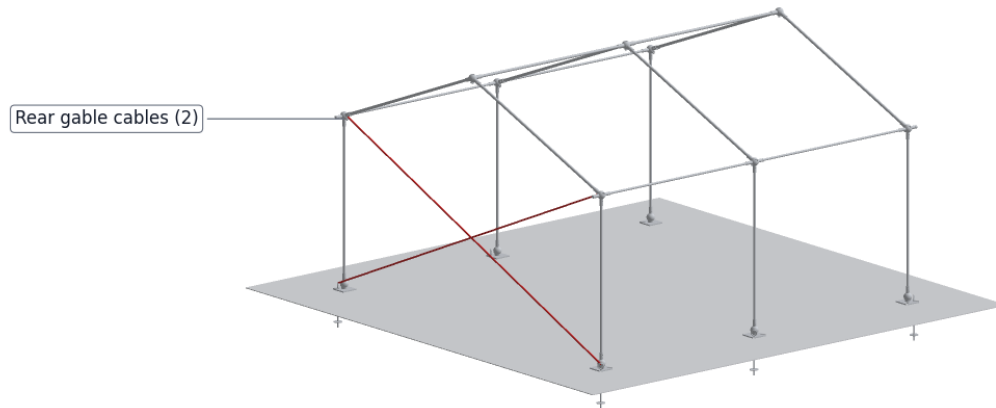
github.com/BoujeeEnjinia1701/snapframe

Check that both floor diagonals are equal within 20 mm; slide feet to suit. At each foot, put a screw anchor through the slot that points away from the middle of the shelter and turn it in by hand, using a spare tube through the eye as a lever, until the eye sits down on the plate across the slot. **Hold point:** safety stop S4.

Step 9: rear gable cables

Step 9: rear gable cables

Each from a rear corner anchor eye to the ring of the opposite eave node; pull snug; tie them where they cross



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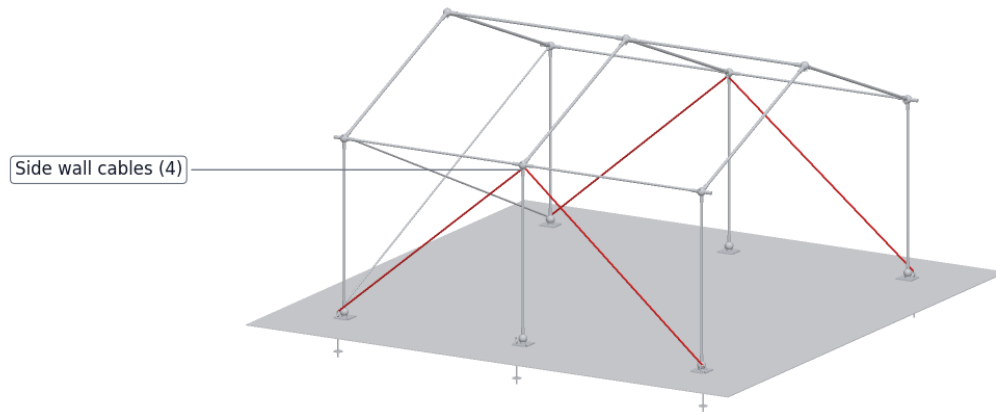
github.com/BoujeeEnjinia1701/snapframe

Seen from behind. Clip each gable cable's hook at the thimble end into a rear corner anchor eye and the other hook into the ring of the opposite rear eave node. Pull both snug with their tensioners, evenly, and tie them together with a cable tie where they cross.

Step 10: side wall cables

Step 10: side wall cables

From each corner anchor eye up to the ring of the middle eave node on that side; pull snug



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

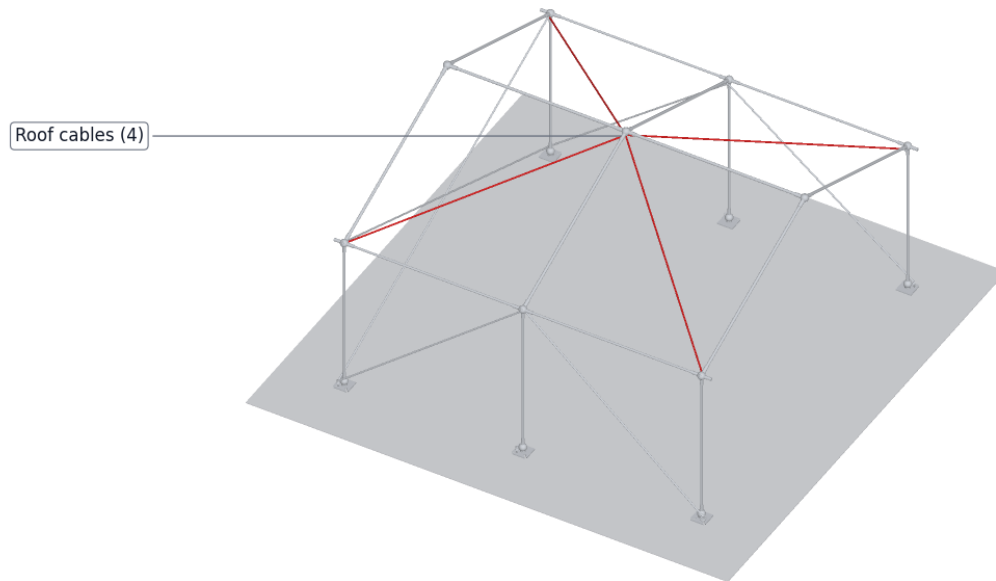
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On each side, clip a wall cable from each corner anchor eye up to the ring of the middle eave node on that side. Pull snug.

Step 11: roof cables

Step 11: roof cables

From the ring of each corner eave node to the ring of the middle ridge node; pull snug, front and back alike



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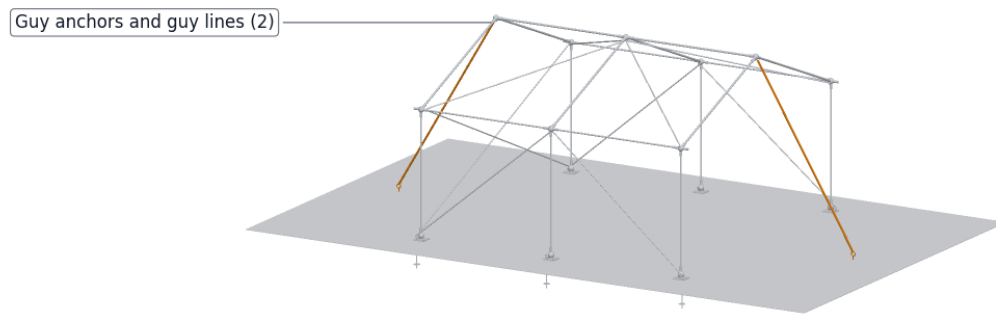
github.com/BoujeeEnjinia1701/snapframe

Clip a roof cable from the ring of each corner eave node to the ring of the middle ridge node. Pull all four snug, front and back alike, so the ridge stays straight.

Step 12: guy anchors and guy lines

Step 12: guy anchors and guy lines

Anchors 1.5 m out from each gable on the ridge line; tie each guy to the end ridge node's ring and tension it



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

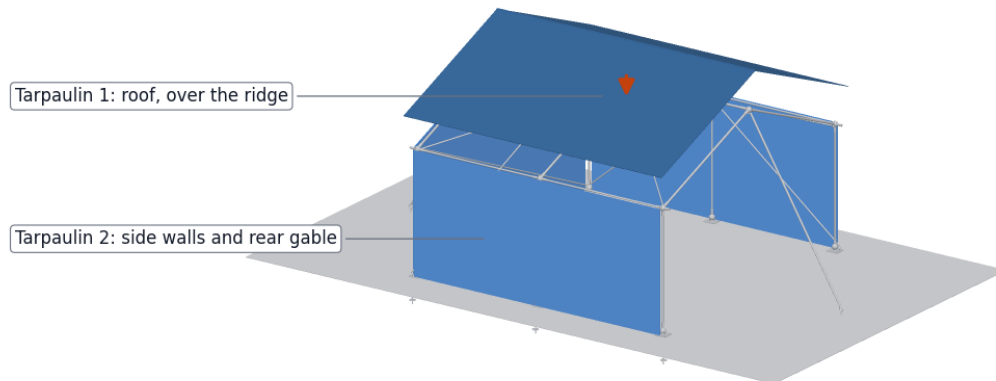
github.com/BoujeeEnjinia1701/snapframe

Turn in a guy anchor on the ridge line 1.5 m out from each gable. Tie each guy line to the ring of its end ridge node and to the guy anchor, and tension it with its slide tensioner. **Hold point:** safety stop S5.

Step 13: tarpaulins

Step 13: tarpaulins (agency stock)

Tie through the eyelets to the tubes, never to the nodes or cables; the front gable stays open



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/snapframe

Throw the first tarpaulin over the ridge as the roof; wrap the second round both side walls and the rear gable. Tie them through their eyelets to the tubes, never to the nodes or the cables. The front gable stays open as the doorway. **Hold point:** safety stop S6.

■ 5. First checks

These are the checks a TRL 4 test report would record; this plan only lists them. Requirement numbers are those of SNF-REQ-001.

Table 2. First checks.

Check	Requirement	How	Pass when
Floor and squareness	R1	Tape the sides and diagonals between the foot centres	4.0 x 4.0 m within 20 mm; diagonals equal within 20 mm
Headroom	R2	Measure from the floor to the underside of the rafters across the span	2.0 m or more over the middle 2.9 m
Joints by hand	R4, R12	Undo and remake one joint of each tube type with a fingertip only	Each button presses flush and the tube comes out and clicks back without tools
Part replacement	R12	Replace one post and one rafter in the standing frame	Each in 5 minutes or less
Anchors	R8	Pull one corner anchor up with a hand scale or a lever and scale	It holds 1.0 kN; record the soil
Cables	R6	Pull each cable to about 300 N with its tensioner and a hand scale	No clip or tensioner slips; the frame stands plumb
Frame stiffness	R6	Push sideways at an eave node and along the ridge with about 100 N	The frame moves a little and returns; no joint opens
Tarpaulin fit	R7	Fit both tarpaulins as step 13	Roof, both side walls and the rear gable closed
Mass and packages	R5, R11	Weigh the tube bundle and the bag	Tube bundle about 28.6 kg, bag about 14.7 kg (both recorded)
Erection time	R4	Two adults who have not seen the kit, with the picture guide	Frame and skin up in 60 minutes or less

■ 6. Safety stops

Stop at each point. Carry on only when everything listed is true.

- **S1. Before cutting and drilling.** Safety glasses on; tube clamped in a vice or vee block, never held by hand under the drill; gloves for handling cut ends.
- **S2. Before printing.** The printer is in a ventilated space or has a filtered enclosure; nylon and other engineering polymers give off fumes while printing.
- **S3. Before a frame is stood up.** Every button clicked and every hitch pin in; two people; wind light (no tarpaulin on the frame).
- **S4. Before any anchor is turned in.** The site is checked for buried cables and pipes with whoever owns them; no anchor within 0.5 m of a marked service.
- **S5. Before the cables and guys are tensioned.** Gloves on; nobody stands in line with a cable while it is being tensioned; tensioners pulled by hand only, never with a bar; guy lines marked with reflective tape.
- **S6. Before the tarpaulins go on.** All anchors, cables and guys fitted and tensioned; wind light. A tarpaulin on an unbraced frame is a sail.
- **S7. Before anyone shelters in it (outside this plan).** The wind rating (about 19.7 m/s, and about 14.7 m/s with wind into the open gable) is posted on the frame; no snow or standing water on the roof; cooking and open flames kept outside; at least 2 m to the next shelter where the site allows; low-voltage lighting only.

■ 7. Tools, skills and workspace

Tools. Pipe cutter for 3/4 in and 1 in conduit, or a hacksaw with a 32 teeth per inch blade; inside reamer and flat file; tape measure, steel rule, marker, a 2 m straight edge (a length of aluminium angle) for the hole line, and a centre punch; bench drill or a drill in a stand, with a vee block and drills of 3, 6 and 8 mm; deburring tool; 3D printer with a bed of at least 230 mm and a hardened nozzle that prints glass- or carbon-filled PA12-class nylon (or a print farm); 16 or 17 mm spanner and a deep socket that fits a 27 mm recess; wire rope cutters and a 10 mm spanner for the rope clips; hand scale to 50 kg; knotted layout cord and six pegs; a step about 0.6 m high; gloves and safety glasses.

Skills. No certified trade is needed. Basic metalwork (measuring, cutting and drilling thin-wall tube), running a 3D printer, bolting, and making up wire rope ends with clips. No electrical work is part of this build.

Workspace. A bench about 2.5 m long for the tubes, or two trestles; a ventilated place for the printer; and a flat, open site at least 8 x 7 m clear of overhead lines and buried services for the first erection, including the guy anchors 1.5 m beyond each gable.

Personal protective equipment. Safety glasses for cutting, drilling and anchor work; cut-resistant gloves for conduit ends and wire rope; hearing protection when sawing; no gloves near a turning drill.

■ 8. Where the numbers come from

- Model and constructability checks: `cad/src/model.py` (`python cad/src/model.py --check` , 203 checks); STEP and STL exports in `cad/step/` and `cad/stl/` .
- Pictures: `cad/src/build_plan_media.py` , using `.kit/build_views.py` ; written to `docs/05-build-plan/` and `cad/drawings/SNF-DWG-101` to `SNF-DWG-109` .
- General arrangement: `cad/drawings/SNF-DWG-001.pdf` , Rev P4.
- Calculations: `docs/04-calcs/01-sizing.md` (SNF-CAL-001 v0.4) and `docs/04-calcs/sizing.py` ; cable bolt and hitch pin, section 6; masses and costs, section 7.
- Bill of materials: `bom/bom.csv` and `bom/bom-notes.md` .
- Decisions: `docs/decisions/0003-design-for-construction.md` (SNF-DDR-003), with SNF-DDR-001 and SNF-DDR-002; open items in `docs/06-design-decisions.md` (SNF-DEC-001).
- Requirements: `docs/03-requirements.md` (SNF-REQ-001 v0.6).